M01 - DATA FOUNDATIONS + ESSENTIAL EXCEL

BEGINNER-FIRST LEARNER BOOK - TEACHING RC2

This book is your teacher. Assume no prior data or Excel knowledge. Do not rush to the practice task. First understand the idea, follow the example, check the expected result, and only then do it alone.

Contents

DE01-01 - What data is: tables, records, fields and data types

DE01-02 - Grain: what one row represents

DE01-03 - Keys, duplicates, NULLs and missing data

DE01-04 - Data quality dimensions and validation thinking

DE01-05 - Meet Excel: workbook, worksheet, cells and formulas

DE01-06 - Enter, save and structure tabular data safely

DE01-07 - Sort and filter without damaging data

DE01-08 - Basic formulas and essential functions

DE01-09 - Relative and absolute references

DE01-10 - Excel Tables and structured data habits

DE01-11 - Practical cleaning and lookup awareness

DE01-12 - PivotTable and Power Query awareness

DE01-13 - Chart choice and basic visual communication

DE01-14 - Reconciliation, sanity checks and source preservation

DE01-15 - Mini workflow: inspect -> calculate -> validate -> explain -> save evidence

DE01-01 - What data is: tables, records, fields and data types

What you are learning

Understand what a table, row, column, cell and data type are before you touch formulas or SQL.

Picture it in your head

Think of a school attendance register. Each student has one row. Name, roll number and attendance are columns. The value where one row and one column meet is a cell.

New words before we use them

Word	Plain-English meaning
Table	A rectangular collection of related data.
Row / record	One observation or event at the table's current level of detail.
Column / field	One attribute stored consistently across rows.
Cell	One value at the intersection of a row and column.
Data type	The kind of value: text, whole number, decimal, date/time, true/false, etc.

Follow with me once

1. Open the workbook sheet **Raw_Sales**.

2. Look only at the header row first: OrderID, OrderDate, CustomerID, CustomerName, Region, Product, Qty, UnitPrice, PaymentMethod.

3. Pick row 2. Read it left to right as one order record.

4. Classify Qty and UnitPrice as numbers, OrderDate as a date-like value, and Region/Product/PaymentMethod as text categories.

What you should see / be able to say

You should be able to point at any value and say both what column it belongs to and what business thing the entire row represents.

Common beginner mistakes

• A value looking like a date does not guarantee Excel stored it as a true date.

• Formatting is appearance; data type is how the value behaves.

• Do not assume every number should be added. OrderID contains digits but is an identifier, not a quantity.

Now you do a small version alone

In Raw_Sales, choose OrderID, Qty, UnitPrice and PaymentMethod. Write whether each is identifier, numeric measure or text category, and explain why.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-02 - Grain: what one row represents

What you are learning

Learn grain - the sentence that says exactly what one row represents.

Picture it in your head

A photo can show one person, one family or one whole crowd. A table also has a level of detail. Grain tells you the level.

New words before we use them

Word	Plain-English meaning
Grain	The business meaning of one row.
Order header	Usually one row per order.
Order line	Usually one row per product line inside an order.
Aggregation	Combining many lower-level rows into fewer summary rows.

Follow with me once

1. Look at Raw_Sales. Ask: if OrderID O1001 appears once, what event does that row describe?

2. Write: **One row represents one order record for one customer/product purchase in this practice data.**

3. Now imagine an order-line table where one order buys three products. That same OrderID could legitimately appear three times because the grain is one order line, not one order.

What you should see / be able to say

Before counting, joining or summarizing, you can say “one row represents ...” in plain English.

Common beginner mistakes

• Saying “grain = customer” just because CustomerID exists.

• Counting rows as orders without proving the table is one row per order.

Now you do a small version alone

For Support_Tickets, write the grain sentence. Then explain why CustomerID may repeat even if TicketID should not.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-03 - Keys, duplicates, NULLs and missing data

What you are learning

Tell legitimate repeats from suspicious duplicates and understand missing values.

Picture it in your head

A person can buy five times, so their customer ID can repeat. But a single receipt number printed twice for the exact same sale is suspicious.

New words before we use them

Word	Plain-English meaning
Key	A column or combination of columns used to identify the row at its intended grain.
Duplicate	A repeated row/key that should be unique at that grain.
NULL / blank	A missing or unknown value.
Zero	A known numeric value equal to nothing - not the same as missing.

Follow with me once

1. In Raw_Sales, find the two O1019 rows. Compare every column. They are identical, so this is suspicious at one-order-record grain.

2. Find CustomerID C003 appearing on more than one order. That is not automatically wrong because one customer can place multiple orders.

3. Find the blank CustomerID and blank PaymentMethod. Treat these as missing-information questions, not as zero.

What you should see / be able to say

You do not delete a repeat until you can explain why it violates the key/grain.

Common beginner mistakes

- Using “Remove Duplicates” before deciding the key.
- Replacing blanks with 0 just to make them disappear.

Now you do a small version alone

In Raw_Sales, write: candidate key, one legitimate repeat, one suspicious duplicate, and two missing-value issues.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-04 - Data quality dimensions and validation thinking

What you are learning

Turn “data quality” from a vague phrase into specific checks.

Picture it in your head

Before accepting a food delivery you might check: did everything arrive, is anything broken, is it the right item, did you receive it on time? Data quality works the same way.

New words before we use them

Word	Plain-English meaning
Completeness	Are required values present?
Validity	Do values obey allowed rules/ranges?
Uniqueness	Are values that should be unique actually unique?
Consistency	Do the same concepts use consistent labels/formats?
Timeliness	Is the data recent enough for the decision?

Follow with me once

1. Completeness: check CustomerID/PaymentMethod blanks.

2. Validity: Qty should normally be positive; Region should belong to the approved set.

3. Uniqueness: OrderID should be unique if grain is one order.

4. Consistency: East and east should not silently become different business regions.

5. Timeliness: an old dataset may be correct but too stale for today's operational decision.

What you should see / be able to say

Every quality claim becomes a concrete test: column + rule + what failure looks like.

Common beginner mistakes

• Writing “check data quality” without a test.

• Confusing uniqueness with “no repeated values anywhere”.

Now you do a small version alone

Using Raw_Sales, write one specific check for each of the five dimensions.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-05 - Meet Excel: workbook, worksheet, cells and formulas

What you are learning

Understand the Excel screen: workbook, worksheet, cell, address, formula bar and formula.

Picture it in your head

A workbook is like a notebook file. Worksheets are the tabs/pages inside it. A cell is one box on a page.

New words before we use them

Word	Plain-English meaning
Workbook	The .xlsx file.
Worksheet	A tab inside the workbook.
Cell address	Column letter + row number, such as D2.
Formula bar	The area that shows the actual value/formula stored in the active cell.
Formula	An instruction beginning with = that Excel calculates.

Follow with me once

1. Open M01_EXCEL_PRACTICE_WORKBOOK_TEACHING_RC2.xlsx.

2. Click the **Formula_Practice** tab.

3. Click cell D2. Notice the Name Box/selected cell address says D2.

4. Type **=B2*C2** and press Enter.

5. Click D2 again. The cell shows a result, while the formula bar shows =B2*C2.

What you should see / be able to say

You understand the difference between what the cell displays and the formula stored behind it.

Common beginner mistakes

• Typing 1700 manually instead of =B2*C2.

• Typing B2*C2 without the leading =.

Now you do a small version alone

Click three different cells and say their addresses aloud. Then write one simple addition formula in an unused cell and undo it.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-06 - Enter, save and structure tabular data safely

What you are learning

Enter and save tabular data without destroying the original source.

Picture it in your head

A raw file is like an original photograph. Edit a copy so you can always return to the untouched original.

New words before we use them

Word	Plain-English meaning
Raw data	Original received data before your transformations.
Working copy	A separate file you are allowed to edit.
Header row	The row containing column names.
Tabular structure	One header row, one observation per row, one attribute per column.

Follow with me once

1. Open the workbook. Use **File > Save As** and create a working copy such as M01_working.xlsx.

2. Keep one header row. Avoid merged cells inside data tables.

3. Do not insert decorative blank rows in the middle of the dataset.

4. Save frequently with Ctrl+S.

What you should see / be able to say

The original workbook remains unchanged, and your working copy contains your experiments.

Common beginner mistakes

• Editing the only copy of the raw file.

• Using two header rows or merged cells as decoration inside source data.

Now you do a small version alone

Create a working copy, rename it clearly, and write the full filename in your evidence notes.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-07 - Sort and filter without damaging data

What you are learning

Use Sort and Filter as viewing/ordering tools without deleting or misaligning data.

Picture it in your head

A filter is like temporarily hiding books that do not match your search; the books are still on the shelf.

New words before we use them

Word	Plain-English meaning
Filter	Temporarily shows rows matching conditions; hidden rows remain in the table.
Sort	Changes row order based on one or more columns.
Whole-row integrity	Values from the same record must move together.

Follow with me once

1. In a working copy of Raw_Sales, click inside the data.

2. Choose **Data > Filter**.

3. Filter Region to East/east and note the visible rows.

4. Clear the filter and confirm the full set returns.

5. For sorting, click inside the table and sort the entire data range by UnitPrice - do not select only UnitPrice cells in isolation.

What you should see / be able to say

Filtering never deletes the hidden rows; sorting keeps OrderID, customer, product, Qty and price aligned on the same record.

Common beginner mistakes

• Deleting hidden rows because they are not visible.

• Sorting one column by itself and scrambling row relationships.

Now you do a small version alone

Filter one PaymentMethod, note the visible count, clear it, and confirm the original row count returns.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-08 - Basic formulas and essential functions

What you are learning

Create basic formulas and use SUM, AVERAGE and COUNT/COUNTA.

Picture it in your head

A formula is a recipe. If the ingredients change, Excel should recalculate the result automatically.

New words before we use them

Word	Plain-English meaning
Cell reference	A pointer to a cell such as B2.
SUM	Adds numeric values.
AVERAGE	Calculates arithmetic mean.
COUNT	Counts numeric cells.
COUNTA	Counts nonblank cells.

Follow with me once

1. Open Formula_Practice. In D2 type =B2*C2.

2. Fill/copy D2 down through the product rows.

3. Manually check Keyboard: 2 x 850 = 1700.

4. Use =SUM(D2:D7) for total Sales.

5. Use =AVERAGE(D2:D7) for average Sales.

6. Use COUNT on a numeric column or COUNTA on a populated text/ID column depending on what you intend to count.

What you should see / be able to say

Changing a Qty or UnitPrice automatically changes the calculated Sales; formulas appear in the formula bar.

Common beginner mistakes

• Typing answers instead of formulas.

• Using COUNT on text IDs and thinking zero means there are no records.

Now you do a small version alone

Complete Formula_Practice and manually validate two rows before trusting the total.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-09 - Relative and absolute references

What you are learning

Understand why A2 moves when copied but H2 stays fixed.

Picture it in your head

A relative reference is like “the house next door” - it changes when you move. An absolute reference is like a full GPS coordinate - it stays fixed.

New words before we use them

Word	Plain-English meaning
Relative reference	A2; changes relative to the formula location when copied.
Absolute reference	H2; row and column are locked.
Mixed reference	$H2 or H$2; only one part is locked.

Follow with me once

1. Open Reference_Practice. Calculate Base Sales in D2 using Qty x UnitPrice.

2. Tax Rate is stored once in H2. In E2 use a formula such as **=D2*(1+H2)**.

3. Copy E2 down. Click E3, E4, E5. D2 changes to D3/D4/D5, but H2 stays exactly H2.

4. Temporarily change H2. Every taxed result should update because all formulas point to the same rate cell. Undo if needed.

What you should see / be able to say

One setting cell controls every row without repeating the percentage in each formula.

Common beginner mistakes

• Using H2 without dollar signs and watching it drift to H3/H4.

• Typing 18% directly in every row formula.

Now you do a small version alone

Complete Independent_Rate using the single rate in H2 and explain what each dollar sign protects.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-10 - Excel Tables and structured data habits

What you are learning

Know the difference between a normal range and an actual Excel Table.

Picture it in your head

A normal range is a group of cells. An Excel Table is a managed object that knows its headers, rows and filters.

New words before we use them

Word	Plain-English meaning
Excel Table	A structured Excel object created with Insert > Table or Ctrl+T.
Header	Named field at the top of a Table column.
Structured behavior	Filters, automatic expansion and formula propagation tied to the Table object.

Follow with me once

1. Work on a clean copy/range. Click inside it.

2. Choose **Insert > Table** (or Ctrl+T).

3. Confirm the full range and tick **My table has headers**.

4. Notice filter arrows appear. Add one new row directly below the Table and see the Table expand.

What you should see / be able to say

The object expands with new rows and keeps explicit headers/filters.

Common beginner mistakes

• Assuming coloured formatting means the data is an Excel Table.

• Creating a Table from an incomplete range that omits columns.

Now you do a small version alone

Convert a small clean range to a Table, add a row, then describe one behavior that proves it is a Table.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-11 - Practical cleaning and lookup awareness

What you are learning

Understand basic cleaning and how XLOOKUP connects related tables using an ID.

Picture it in your head

A lookup is like using a roll number to find a student's house from a separate register. The ID is the matching key.

New words before we use them

Word	Plain-English meaning
Cleaning	Making values consistent/usable without changing their business meaning.
Lookup value	The ID you are trying to find.
Lookup array	Where Excel searches for that ID.
Return array	The column from which Excel returns the answer.
Not found	The explicit result when no match exists.

Follow with me once

1. Notice Region values East, east and West with trailing space in Raw_Sales. These are consistency problems.

2. Open Customer_Lookup. CustomerID is the linking field; Tier is the value you want to bring back.

3. In Independent_Lookup, use XLOOKUP with CustomerID as the lookup value, Customer_Lookup!CustomerID as search range, and Tier as return range.

4. Handle C999 explicitly with a not-found message rather than silently assuming a tier.

What you should see / be able to say

Known IDs return the same Tier shown in Customer_Lookup; C999 is clearly flagged missing.

Common beginner mistakes

- Matching by CustomerName when CustomerID is the intended key.
- Treating unmatched IDs as blank and forgetting to investigate them.

Now you do a small version alone

Complete Independent_Lookup, spot-check two known customers, and explain the four pieces of XLOOKUP in plain English.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-12 - PivotTable and Power Query awareness

What you are learning

Understand what PivotTables and Power Query are for without trying to master them yet.

Picture it in your head

PivotTable is a fast summary machine. Power Query is a recorded cleaning/import recipe you can replay.

New words before we use them

Word	Plain-English meaning
PivotTable	Interactive summary built from source data.
Rows area	Categories listed vertically.
Values area	Measures summarized, e.g., Sum of Sales.
Power Query	Excel's repeatable import/transform tool.
Refresh	Run the saved import/transform logic again on updated data.

Follow with me once

1. Pivot: click Pivot_Source > Insert > PivotTable > New Worksheet. Put Region in Rows and Sales in Values.

2. Check that Values says **Sum of Sales**, not Count of Sales.

3. Power Query awareness: inspect PowerQuery_Source. Find leading/trailing spaces, inconsistent case, mixed date formats and Amount stored as text with comma.

4. The point is not to hand-fix every cell. A Power Query workflow can record trim/clean/type/standardization steps and refresh them later.

What you should see / be able to say

You can answer which Region has the highest sales and describe Power Query as repeatable transformation rather than “another worksheet”.

Common beginner mistakes

• Building a fancy dashboard before understanding the summary.

• Using Count of Sales by accident.

• Manually cleaning source values one-by-one and calling that a repeatable pipeline.

Now you do a small version alone

Create the simple PivotTable, reconcile its grand total to source SUM, then write 2-3 sentences explaining Power Query.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-13 - Chart choice and basic visual communication

What you are learning

Choose a simple chart that matches the question and avoid decorative charts that hide the message.

Picture it in your head

A chart is a sentence in picture form. Choose the picture that makes the comparison easiest to read.

New words before we use them

Word	Plain-English meaning
Bar/column chart	Good for comparing categories.
Line chart	Good for trends over ordered time.
Pie chart	Limited use; difficult when many categories/slices exist.
Axis	Scale along horizontal/vertical direction.
Title	Should state what is being compared, not just “Chart 1”.

Follow with me once

1. Use the regional summary from your PivotTable.

2. For comparing East/West/North/South, choose a bar or column chart.

3. Give it a clear title such as “Sales by Region”.

4. Check that the largest bar visually matches the largest numeric total.

What you should see / be able to say

A reader can answer the business question quickly without decoding unnecessary colours or 3-D effects.

Common beginner mistakes

• Using a line chart for unordered categories.

• Adding 3-D effects that distort comparisons.

• Making the title vague.

Now you do a small version alone

Create one simple regional comparison chart and write one sentence explaining why that chart type fits the question.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-14 - Reconciliation, sanity checks and source preservation

What you are learning

Learn to distrust even plausible-looking results until you reconcile them to an independent check.

Picture it in your head

If a shop cashier says the total is 3810, you may still count the item totals separately. Reconciliation is that second route to the answer.

New words before we use them

Word	Plain-English meaning
Reconciliation	Comparing the result to an independent total/count/control.
Source preservation	Keeping raw evidence unchanged.
Sanity check	A quick reasonableness check for impossible or surprising values.
Control total	A known count/amount used to verify processing.

Follow with me once

1. Before transformations, record source row count and any important total.

2. After filtering/cleaning/calculating, recompute the total another way.

3. If Pivot grand total differs from SUM of source Sales, stop and find the earliest step where rows/values diverged.

4. Do not “fix the final number” manually; repair the logic.

What you should see / be able to say

You can show both the result and one independent check that supports it.

Common beginner mistakes

• Trusting a pretty PivotTable because it looks professional.

• Changing raw data to force agreement.

Now you do a small version alone

For Pivot_Source, compare Pivot grand total with =SUM(E2:E19). Record both and explain whether they reconcile.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

DE01-15 - Mini workflow: inspect -> calculate -> validate -> explain -> save evidence

What you are learning

Combine the module into a repeatable beginner workflow: inspect -> calculate -> validate -> explain -> save evidence.

Picture it in your head

A good analyst is not someone who can press many Excel buttons. It is someone who can explain what data they received, what they changed, what result they produced, and why they trust it.

New words before we use them

Word	Plain-English meaning
Inspect	Understand columns, grain, keys and quality before calculating.
Calculate	Apply formulas/summaries that answer the question.
Validate	Use an independent check.
Explain	State result and reasoning in plain language.
Evidence	Saved workbook, note, screenshot, formula or output proving what you actually did.

Follow with me once

1. Read the business question first.

2. State the table grain and candidate key.

3. Preserve raw source and work on a copy.

4. Perform only the transformations/calculations needed.

5. Validate with row counts, control totals or spot checks.

6. Explain the answer, one limitation and one possible failure mode.

7. Save the workbook and note where the evidence lives.

What you should see / be able to say

Someone else can reproduce and audit your work instead of trusting an unexplained final number.

Common beginner mistakes

• Starting with a chart/formula before understanding the data.

• Saving only screenshots and losing the actual workbook/formulas.

• Reporting “done” with no validation.

Now you do a small version alone

Write a 5-8 sentence handoff for one M01 practice task: source, grain, what you changed, result, validation, limitation.

Do not mark this lesson mastered just because you read it.

Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

End-of-module reminder

The short Quick Reference is for retrieval after learning. The Guided Labs are for follow-along practice. Independent Practice is for proving you can work without copying. None of those should replace the teaching above.